Open-Source Development

ITIC2262

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2. Describe open-source software development concepts

* Software Development Process

Intended Learning Outcomes

* After the completion of this module, the student will be able to Describe open-source software development concepts..
  - Explain the software development process.
  - Describe the Software Development Life Cycle (SDLC).
  - Understand the stages of open-source software development.
  - Identify the roles of contributors in OSS projects.
  - Explain collaborative development using Git and GitHub.

What is Software Development

* **Definition**

Software development is the systematic process of designing, creating, testing, deploying, and maintaining software applications to solve user problems.

It includes:

- Requirement gathering
- Software design
- Programming
- Testing
- Deployment
- Maintenance

What is Open Source Software Development

Open Source Software Development is a collaborative software development approach where the source code is publicly available.

* Anyone can:
  - View the source code
  - Modify it
  - Improve it
  - Share improvements
  - Report issues
  - Contribute documentation

* Examples:
  - Linux
  - Mozilla Firefox
  - LibreOffice
  - WordPress

Software Development Life Cycle (SDLC)

* **Main Phases**

1. Planning
2. Requirements Analysis
3. System Design
4. Development (Coding)
5. Testing
6. Deployment
7. Maintenance

Planning

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Requirements

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Design

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Development

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Testing

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Deployment

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Maintenance

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Continuous Improvement

Phase 1: Planning

* **Planning identifies:**
  - Project goals
  - Target users
  - Project scope
  - Budget
  - Timeline
  - Development tools

* **Deliverables:**
  - Project proposal
  - Development roadmap

Phase 2: Requirements Analysis

* Developers gather user needs.
* Types of requirements.

* **Functional Requirements**
  - Login
  - Search
  - Reports

* **Non-functional Requirements**
  - Security
  - Performance
  - Reliability
  - Usability

Phase 3: System Design

* **Developers prepare:**
  - System architecture
  - Database design
  - User Interface (UI)
  - APIs
  - Module structure

* **Common tools:**
  - UML diagrams
  - Flowcharts
  - Wireframes

Phase 4: Development (Coding)

* **Programmers write code.**
  * Activities include:
  - Writing source code
  - Code review
  - Version control
  - Documentation

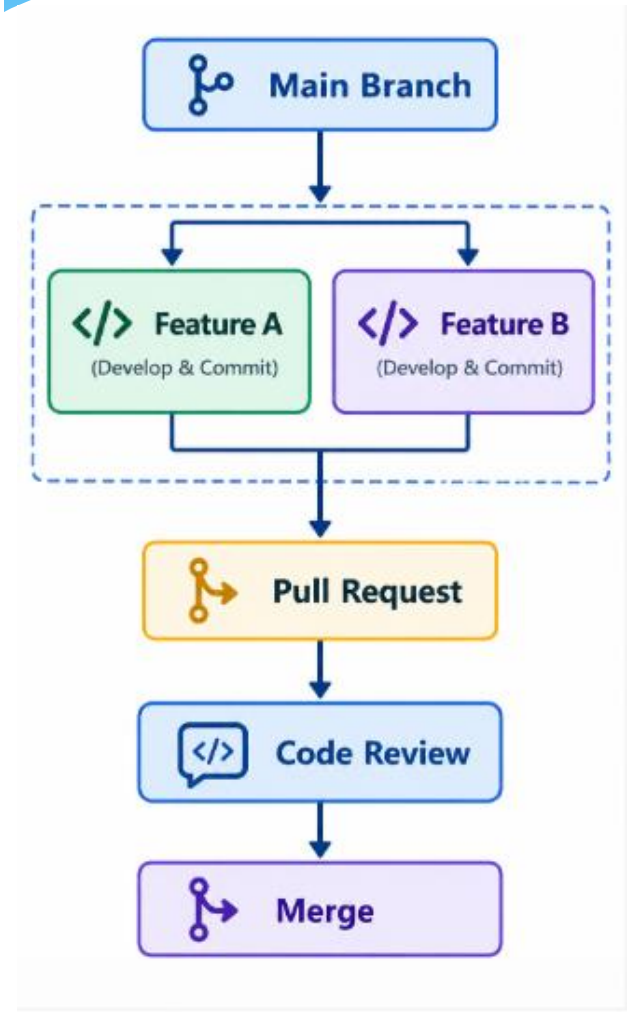
* **Common programming languages:**
  - Python
  - Java
  - JavaScript
  - C++
  - PHP

Version Control in Open Source

* **Version control helps developers:**
  - Track changes
  - Restore previous versions
  - Collaborate safely
  - Manage multiple contributors

* **Popular tools**
  - Git
  - GitHub
  - GitLab
  - Bitbucket

Branching Workflow



❖ **Benefits**

- Safe development
- Parallel work
- Easy collaboration

Phase 5: Testing

* Testing ensures software quality

* **Types**
  - Unit Testing
  - Integration Testing
  - System Testing
  - Acceptance Testing
  - Regression Testing

* **Objectives**
  - Find bugs
  - Improve reliability
  - Ensure correctness

Phase 6: Deployment

* Deployment means releasing software to users.

Methods

- Manual deployment
- Automated deployment
- Continuous Deployment

* Example

Developer → GitHub → Build → Testing → Release → Users

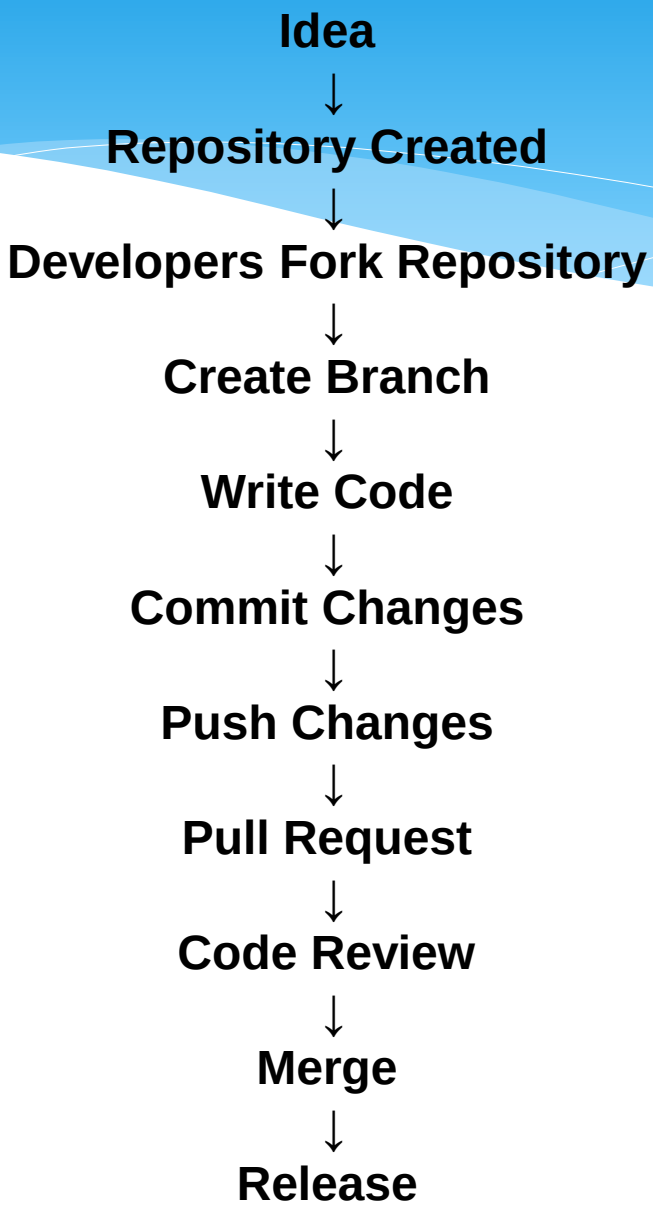
Phase 7: Maintenance

After deployment:

* **Developers**
  - Fix bugs
  - Improve performance
  - Add new features
  - Update documentation
  - Improve security

* Maintenance continues throughout the software lifecycle.

Open Source Development Workflow



Roles in Open Source Projects

Role	Responsibility
Project Owner	Manages the project
Maintainer	Reviews and merges code
Developer	Writes code
Tester	Finds bugs
Documentation Writer	Writes guides
Community Member	Reports issues and suggests improvements

Advantages of Open Source Development Process

- Global collaboration
- Faster innovation
- Transparent development
- Better software quality
- Continuous improvements
- Cost-effective
- Community support

Challenges

- Large number of contributors
- Code conflicts
- Security vulnerabilities
- Documentation quality
- Project management
- Sustainability
- Quality assurance

Case Study

* **Linux Development Process**
  1. Community reports issues
  2. Developers submit patches
  3. Maintainers review code
  4. Testing performed
  5. Code merged
  6. New kernel released

* This collaborative model enables continuous improvement and supports one of the world's most widely used operating systems.

Questions

1. What is the Software Development Life Cycle (SDLC)?
2. What are the seven main phases of software development?
3. Why is version control important in open-source projects?
4. What is the purpose of a pull request?
5. What roles contribute to an open-source software project?

THANK YOU